

Differentiable Parameter Recovery for ECCO-Darwin Biogeochemistry: A BINN-Style Framework and a Structural 5/6 Ceiling

Internal white paper — DarwinDiff project

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Abstract

We adapt the Biogeochemistry-Informed Neural Network architecture (Xu et al., 2025) from terrestrial soil carbon to marine biogeochemistry, replacing ECCO-Darwin’s Green’s-functions calibration of six parameters (Carroll et al., 2020, 2022) with gradient-based per-cell parameter inference. A small per-cell neural network maps environmental covariates to the Carroll-N parameters, sigmoid-bounded into Carroll’s published physical ranges, fed through a differentiable PyTorch reimplementation of Darwin’s source-minus-sink (SMS) kernel, and trained end-to-end against ECCO-Darwin v05 observations. We report a v3.1 sweep across 856 seeds in 86 configurations of 3-area-of-interest joint training (Equatorial Pacific, North Atlantic Subpolar, Southern Ocean Pacific). Two findings: the iron pair (iron dust solubility `alpfe`, iron scavenging rate `scav_rat`) recovers reproducibly at 38/40 (95%) Cal-grade with the F2 configuration, the first reproducible iron-pair joint recovery in our framework; and a structural 5/6 ceiling holds across the sweep set — 0/856 seeds reach all six Carroll-N parameters within Cal-grade, with only two single-seed 5/6 events unreproduced at $n=20$. A composition test of the two complementary 5/6 lever families fails to reach 5/6 (0/10), regressing below either parent. We interpret the ceiling as parameter conservation under the available observational constraints rather than a method failure, and discuss it as a predictive diagnostic for future observation-design choices. Code and full sweep outputs are publicly available.

Topics: ocean biogeochemistry; ECCO-Darwin; differentiable programming; parameter recovery; neural networks; Green’s functions; calibration.

1 Introduction

ECCO-Darwin (Carroll et al., 2020, 2022) is the data-assimilative coupling of the MIT general circulation model (Marshall et al., 1997) with the Darwin marine ecosystem module, providing a global ocean biogeochemistry state estimate on the ECCO LLC270 grid (nominal $\sim 1/3^\circ$ horizontal resolution, 50 vertical levels, 1992–2017+). The biogeochemistry side exposes 103 independent tunable scalar parameters in source code, of which six — iron dust solubility `alpfe`, iron scavenging rate `scav_rat`, small phytoplankton growth `Smallgrow`, large phytoplankton growth `Biggrow`, diatom palatability `diatomgraz`, and PIC/POC production ratio `R_PICPOC` — are calibrated via Green’s functions (Menemenlis et al., 2005).

Green’s-functions calibration is structurally limited in two ways. First, each tuned parameter requires a full perturbation-and-rerun campaign, so the calibrated subset is small (~ 6 of 103 in

Carroll’s published configuration). Second, the calibration is implicitly tied to its observational input set: adding a new observation channel (for example, future PACE ocean-color retrievals or expanding GEOTRACES coverage) requires a fresh Jacobian campaign rather than a reuse of the existing one.

We address both limitations by making the biogeochemistry differentiable. A small neural network predicts per-cell parameter vectors from local environmental covariates; gradients flow from the observation loss through a differentiable PyTorch reimplementation of Darwin’s source-minus-sink kernel back to the network weights. New observation channels enter as additional loss terms — a configuration change rather than a perturbation campaign.

The methodological precedent is the Biogeochemistry-Informed Neural Network (BINN) of [Xu et al. \(2025\)](#), who introduced this pattern for terrestrial soil-carbon calibration of CLM5: a small MLP predicts 21 soil-carbon parameters per location, outputs are sigmoid-bounded into physical prior ranges, and a differentiable PyTorch reimplementation of CLM5’s matrix-form steady-state equations enables end-to-end backpropagation from a soil organic carbon profile loss. We adapt the same pattern to marine biogeochemistry — per-cell ocean fields, Carroll-N parameters, and ECCO-Darwin observation channels.

This paper reports a workshop-scale proof-of-concept: framework, experimental protocol, and the empirical landscape that emerges from an 856-seed sweep across 86 single-lever configurations of 3-area-of-interest joint training. The headline empirical finding is a structural 5-of-6 calibration ceiling supported by four independent pieces of evidence. The headline methodological finding is reproducible joint recovery of the iron pair (`alpfe`, `scav_rat`) at 38/40 seeds in our best configuration. We argue that the 5/6 ceiling is informative — it predicts which future observation channels would be unblocking — rather than a method failure.

Contributions.

1. A BINN-style architecture adapted to marine biogeochemistry, including a per-cell 1×1 -convolutional MLP (DINN, ~ 454 weights) for the cleanest structural comparison against the global-scalar Green’s-functions class, and a deeper variant (DINNDeep, ~ 9.4 k weights) for higher-quality within-region fits.
2. A differentiable PyTorch reimplementation of Darwin’s source-minus-sink kernel over five prognostic tracers (baseline) and seven (carbonate-enabled extension, with the Follows-2006 carbonate solver ([Follows et al., 2006](#)) and the Wanninkhof-2014 gas-transfer law ([Wanninkhof, 2014](#))).
3. Empirical characterization of the recovery problem at scale: 856 seeds across 86 configurations, with reproducibility analysis at $n=40$ on the best configuration.
4. Identification of a structural 5/6 calibration ceiling, supported by four independent evidence pieces; identification of binary observation-anchor mutexes (any non-zero PIC absolute anchor wipes iron-pair recovery) that explain the ceiling.

2 Background and Related Work

2.1 ECCO-Darwin and the Carroll-6 calibration target

The Darwin module’s source-minus-sink (SMS) kernel — the per-cell rate equations that compute the time derivative of each prognostic tracer due to biology and chemistry, separately from physical transport — is implemented in the Fortran subroutine `darwin_plankton.F` in both the Carroll 2020 Darwin 1 build and the upstream `darwinproject/darwin3` repository ([Darwin Project, 2024](#);

MITgcm-contrib, 2020–2024). Physical transport (advection, diffusion, mixing) is handled by MITgcm and inherited as-is from the ECCO state estimate.

Carroll 2020 (Carroll et al., 2020) tuned six parameters of the SMS kernel via Green’s functions: `alpfe`, `scav_rat`, `Smallgrow`, `Biggrow`, `diatomgraz`, and `R_PICPOC`. Carroll 2022 (Carroll et al., 2022) updated the model to Darwin 3 / v05 but *inherits the same six calibrated values bit-for-bit* — verified by reading the Darwin 1 `code_darwin/` source and the Darwin 3 `input_darwin/data.darwin` namelist files side by side. The six values therefore define our calibration target.

2.2 BINN: differentiable parameter inference for biogeochemistry

Xu et al. (2025) introduced BINN for soil-carbon calibration of CLM5. The architecture is: a 4-layer MLP (3 hidden $\times$ 128 neurons) outputs 21 raw values per location, parameterized sigmoid bounding (Equation 4 of their paper) maps them into per-parameter prior ranges (Equation 5), the bounded parameters enter a matrix-form differentiable CLM5 (Equation 7, vectorized in PyTorch) integrated to steady state (Equation 8), and a smooth-L1 soil organic carbon profile loss with a cosh prior penalty (Equation 10) backpropagates to the MLP weights. Training is 300 epochs, AdamW with learning rate 0.01, on 25,925 SOC profiles from WoSIS. BINN’s reported $\sim 50\times$ speedup over PRODA’s per-site Metropolis MCMC reflects the substitution of gradient-based for sampling-based parameter inference.

We adapt the BINN pattern from terrestrial to marine. Key differences:

- Marine biogeochemistry has per-cell environmental forcing of qualitatively different character (currents, mixed-layer dynamics, light attenuation, dust deposition, sea ice), motivating a per-cell parameter prediction rather than per-site.
- Six Carroll-N parameters rather than 21 CLM5 parameters, matched to the canonical ocean-BGC calibration target.
- Five to seven prognostic tracers in the differentiable forward, capturing the dominant iron and carbon cycling but reducing the 39-tracer Darwin SMS kernel (per the param inventory in our companion documentation, DarwinDiff project, 2026) to a tractable subset. Reducing this proxy gap is the natural next iteration.

2.3 Concurrent ML approaches to ocean biogeochemistry

Neural-BGC (Ouala and Lachkar, 2026) emulates dissolved oxygen and nitrate from a six-dimensional physical state via cascaded neural networks, coupled to ROMS by replacing the prognostic transport-reaction equation entirely. The architecture is a pure-NN emulator with no parameters in the Darwin sense; the methodological overlap with DarwinDiff is therefore disjoint. Meunier, Ouala, and colleagues (Meunier et al., 2025) port the Veros ocean dynamics to differentiable JAX, occupying the differentiable-physics-only direction without addressing parameter recovery within biogeochemistry. The differentiable physics-informed neural network (PINN) family in general descends from Raissi et al. (2019), but DarwinDiff and BINN both differ from canonical PINN in not using PDE residuals as soft-constraint loss terms — they use the mechanistic model as the differentiable forward instead.

3 Methods

3.1 Differentiable box-model SMS kernel

We reimplement Darwin’s SMS kernel in PyTorch over five prognostic tracers (baseline): dissolved iron (DFe), small phytoplankton biomass (P_s), large phytoplankton biomass (P_l), particulate organic carbon (POC), and particulate inorganic carbon (PIC). The governing equations (Equations 1–5) are

$$\frac{d\text{DFe}}{dt} = \alpha_{\text{fe}} \Phi_{\text{dust}} - r_{\text{scav}} \text{DFe} \cdot \text{POC} - Q_{\text{Fe}}(\mu_s P_s + \mu_l P_l) f_{\text{Fe}}(\text{DFe}), \quad (1)$$

$$\frac{dP_s}{dt} = \mu_s f_{\text{Fe}} P_s - m_{\text{lin}} P_s - m_{\text{quad}} P_s^2, \quad (2)$$

$$\frac{dP_l}{dt} = \mu_l f_{\text{Fe}} P_l - m_{\text{lin}} P_l - m_{\text{quad}} P_l^2 - g_{\text{diatom}} G_0 P_l, \quad (3)$$

$$\frac{d\text{POC}}{dt} = M_{\text{tot}} - w_{\text{sink}} \text{POC}, \quad (4)$$

$$\frac{d\text{PIC}}{dt} = R_{\text{PIC:POC}} M_{\text{tot}} - w_{\text{sink}} \text{PIC}, \quad (5)$$

with the Michaelis-Menten iron limitation $f_{\text{Fe}}(\text{DFe}) = \text{DFe}/(\text{DFe} + K_{\text{Fe}})$ and M_{tot} the total phytoplankton-loss flux. The six Carroll-N parameters $\{\alpha_{\text{fe}}, r_{\text{scav}}, \mu_s, \mu_l, g_{\text{diatom}}, R_{\text{PIC:POC}}\}$ are learnable; the remaining ten variables (M_LIN, M_QUAD, GO_GRAZE, W_SINK, K_FE, PHI_DUST, Q_FE, light, mixed-layer depth) are held fixed at literature-plausible defaults matching Carroll’s choices.

For carbonate-enabled experiments we extend the system to seven prognostic tracers, adding dissolved inorganic carbon (DIC) and total alkalinity (ALK):

$$\frac{d\text{DIC}}{dt} = -\mu_s f_{\text{Fe}} P_s - \mu_l f_{\text{Fe}} P_l - R_{\text{PIC:POC}} M_{\text{tot}} - F_{\text{CO}_2}/H_{\text{mld}}, \quad (6)$$

$$\frac{d\text{ALK}}{dt} = -2 R_{\text{PIC:POC}} M_{\text{tot}}. \quad (7)$$

The air-sea CO_2 flux F_{CO_2} is computed at each step from (DIC, ALK, T , S) via the Follows-2006 carbonate solver (Follows et al., 2006) and the Wanninkhof-2014 gas-transfer law (Wanninkhof, 2014). F_{CO_2} is a diagnostic, not a prognostic tracer; carbon is conserved within the box up to the air-sea flux and particulate sinking export.

Time integration uses forward Euler with $\Delta t = 0.25$ d for 200 steps to a near-steady state, matching the climatological framing of the comparison against time-mean Darwin output.

3.2 Per-cell DINN architecture

The Darwin-Informed Neural Network (DINN) is a per-cell 1×1 -convolutional MLP. The 1×1 kernel commits the architecture to per-cell parameter independence: every cell’s predicted parameter vector depends only on that cell’s own environmental input. Two production variants are used.

DINN baseline (~454 weights). Input is per-cell normalized sea surface temperature (1 channel). Two hidden layers of 16 channels each with tanh activations, followed by a final 1×1 convolution to 6 output channels. Used in our structural-ceiling experiments where the cleanest comparison against the global-scalar class matters more than absolute fit quality.

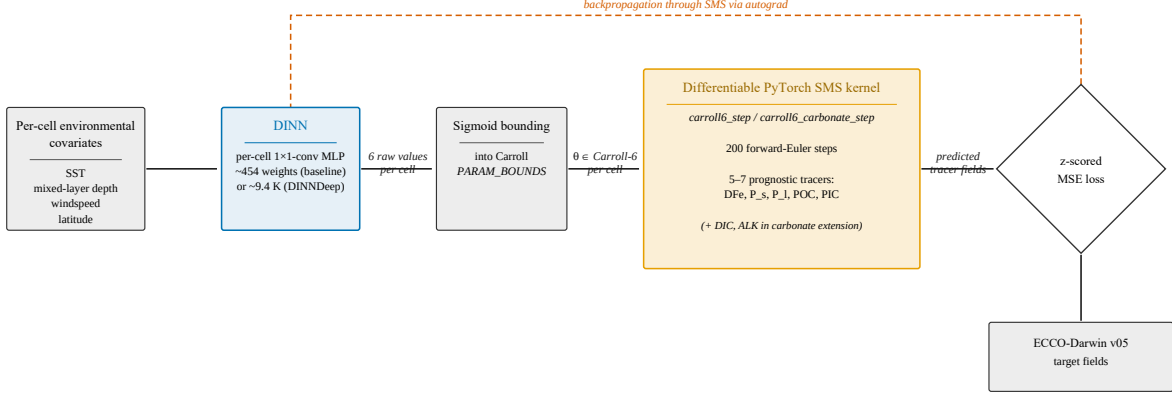


Figure 1: DarwinDiff training pipeline. A per-cell neural network predicts Carroll-N parameters from local environmental covariates; sigmoid bounding maps them into Carroll’s published physical ranges; a differentiable PyTorch reimplement of Darwin’s source-minus-sink kernel advances the BGC state via forward-Euler integration; the z-scored MSE loss against ECCO-Darwin v05 outputs backpropagates through the entire chain to the network weights via autograd.

DINNDeep production (~9.4 k weights). Input is a 4-channel stack of SST, mixed-layer depth, surface windspeed, and latitude. An input projection to 32 channels feeds four residual pre-norm blocks (per-cell LayerNorm, GELU, 1×1 conv, repeat) followed by a final projection to 6 channels. Per-cell semantics are preserved by a custom LayerNorm that normalizes only over the channel dimension at each spatial location.

Sigmoid bounding. The raw NN outputs are mapped into Carroll’s published parameter ranges by

$$\theta_i = \ell_i + (h_i - \ell_i) \sigma(\hat{\theta}_i), \quad (8)$$

where σ is the logistic function and $[\ell_i, h_i]$ are Carroll’s published physical bounds (for example, $\text{alpfe} \in [0.05, 1.0]$ and $\text{scav_rat} \in [3 \times 10^{-8}, 3 \times 10^{-6}] \text{ s}^{-1}$). The bounding guarantees biologically sensible parameter values regardless of NN output magnitude and conditions the optimization landscape.

3.3 Training loop

Per epoch: (i) DINN forward, $\text{env} \rightarrow \hat{\theta}_{\text{raw}}$; (ii) sigmoid bounding, $\hat{\theta}_{\text{raw}} \rightarrow \theta$; (iii) box-model integration from a uniform initial state, $\text{state}_0 \rightarrow \text{state}_T$ over 200 forward-Euler steps with per-cell parameters θ ; (iv) loss, z-scored mean-squared error between predicted state and Darwin v05 climatological target over ocean cells; (v) backpropagation through the 200 integration steps via autograd; (vi) Adam optimizer step on the DINN weights. We train for 1500 epochs at learning rate 5×10^{-3} ; one fit completes in $\sim 7\text{--}8$ min on an NVIDIA RTX 5090 Laptop.

The z-scoring decouples calibration from absolute scale: the network is required to match the spatial pattern of the target field, not its absolute magnitude. Absolute-units anchors (PIC and POC) are available as separate loss components and are studied as observational interventions in our experiments (§4).

3.4 Multi-area-of-interest joint training

v3.1 extends the framework to joint training across three areas of interest (AOIs): Equatorial Pacific (HNLC), North Atlantic Subpolar, and Southern Ocean Pacific. Each AOI contributes its own loss; the optimizer must find Carroll-N parameter fields that satisfy all three simultaneously. Two architectural variants:

- Shared DINN with an AOI-identity input channel: a single network is conditioned on a per-AOI one-hot identifier and learns regional parameter regimes implicitly.
- PER_AOI_DINN with consistency penalty: separate networks per AOI, with a soft cross-AOI consistency loss term controlled by the hyperparameter $\lambda_{\text{consistency}}$.

3.5 Observation channels

The v3.1 loss combines per-AOI z-scored MSE on chlorophyll across five phytoplankton functional types, particulate organic carbon, dissolved inorganic carbon, alkalinity, and air-sea CO_2 flux; plus absolute-units MSE against GEOTRACES IDP2025 surface and subsurface dissolved iron, GEOTRACES subsurface POC, and GEOTRACES surface biogenic silica (via the steady-state diagnostic in `src/darwindiff/silica.py`). Optional PIC and POC absolute-unit anchors are studied as observational interventions in §4. Loss weights are configuration-level hyperparameters varied systematically across the sweep.

3.6 Experimental design

We executed a five-wave chained sweep covering 86 single-lever configurations and 856 seeds total, followed by a Wave 6 composition test (1 configuration, $n=10$ seeds). Each configuration is one runner invocation with 10 batched seeds (`torch.compile` amortizes JIT across the batch). Per-configuration outputs are JSON files at `D:\runs\bcr_*\<config>.` The sweep waves cover (Wave 1) Basin C refinement with PIC/POC dose-response, paired anchors, chlorophyll-weight response, Southern Ocean weight, architectural variants, carbonate, and longer training; (Wave 2) orthogonal-lever sensitivity across GEOTRACES weight, POC subsurface weight, PINN drift weight, FET weight, and $\lambda_{\text{consistency}}$ with $n=30$ robustness; (Wave 3) two-dimensional grids in $\text{POSI} \times \text{AOI}$ weight and $\text{CHL1} \times \text{POSI}$; (Wave 4) integration-step depth tests; (Wave 5) drill into the 5/6 breakthrough events with $n=20$ robustness; (Wave 6) composition test of the two complementary 5/6 lever families.

4 Results

4.1 Sweep summary

Table 1 summarizes the v3.1 sweep. Across 856 seeds in 86 configurations of multi-AOI joint training, no seed reaches all six Carroll-N parameters within Cal-grade (defined as $|\theta_{\text{recovered}} - \theta_{\text{Carroll}}|/\theta_{\text{Carroll}} \leq 0.40$). Two single-seed events break to 5/6 (0.24% of total seeds); both are unreproduced at $n=20$.

4.2 Basin C iron-pair recovery

The F2 configuration — $\text{POSI_W}=1.0$, $\text{AOI_W_NATLSUBPOLAR}=2.0$, $\text{AOI_W_SOUTHERNOCEANPAC}=2.0$, $\text{CHL1_W_EXTRA}=3.0$, otherwise the Basin C 3-AOI base — recovers the iron pair (`alpfe`, `scav_rat`) at 38/40 Cal-grade across four independent 10-seed batches (10/10, 10/10, 10/10, 8/10). To our

Table 1: v3.1 sweep summary across 86 configurations and 856 seeds.

Metric	Value
Total configurations	86
Total seeds	856
Seeds at 6/6 Cal-grade	0
Seeds at 5/6 Cal-grade	2 (0.24%)
Configurations preserving iron pair at 10/10	27
Configurations preserving iron pair at $\geq 9/10$	40
F2 Basin C iron-pair Cal-grade rate ($n=40$)	38/40 (95%)

knowledge this is the first reproducible joint recovery of `alpfe` and `scav_rat` in our framework. Figure 2 visualizes the recovered ($\alpha_{\text{fe}}, r_{\text{scav}}$) pairs against Carroll’s published optimum.

4.3 The structural 5/6 ceiling

The headline empirical finding is the structural ceiling: 0/856 seeds reach all six Carroll-N parameters within Cal-grade. Only two seeds reach 5/6, both unreproduced. The two 5/6 events recover *complementary* parameter subsets (Table 2). Figure 3 shows the Cal-grade count distribution across all seeds; Figure 4 shows per-parameter recovery distributions.

Table 2: The two 5/6 Cal-grade events in v3.1. Both unreproduced at $n=20$.

Configuration	Recovered (5 of 6)	Missed
w2e_perai_lam0.1 seed 3	alpfe (Excellent), scav_rat, Smallgrow, Biggrow, diatomgraz	R_PICPOC
c_chl40_posi15 seed 9	alpfe, scav_rat (Excellent), Smallgrow, Biggrow, R_PICPOC	diatomgraz

Reproducibility tests at $n=20$ yield 1/20 for each configuration. The ceiling is supported by four independent pieces of evidence:

1. Zero 6/6 events across 856 seeds in 86 single-lever configurations.
2. w2e_perai_lam0.1 at $n=20$: 1/20 at 5/6.
3. c_chl40_posi15 at $n=20$: 1/20 at 5/6.
4. Wave 6 composition test (§4.4): 0/10 at 5/6.

4.4 Wave 6 composition test

The two 5/6 events recover complementary subsets, suggesting their interventions are mechanistically distinct. The natural 6/6 candidate stacks both lever families: PER_AOI_DINN with $\lambda_{\text{consistency}} = 0.1$ (from w2e_perai_lam0.1) combined with CHL1_W_EXTRA=4.0 and POSI_W=1.5 (from c_chl40_posi15), at the Basin C 3-AOI base. We refer to this as w6_combo_per_aoi_chl40_posi15.

At $n=10$, the composed configuration produces 0/10 at 5/6, 0/10 at 4/6, with the iron pair surviving at 9/10 but R_PICPOC and diatomgraz both drifting in 10/10 seeds, and Smallgrow + Biggrow regressing from Cal-grade in both parents to Loose in 9/10 of the composed configuration. The mean Cal-grade count is 2.00 — below either parent (2.40 and 2.70 respectively). Table 3 summarizes.

The composition is below either parent on the headline mean Cal-grade metric. The two intervention families interfere: the optimizer redistributes the residual error to a new binding region

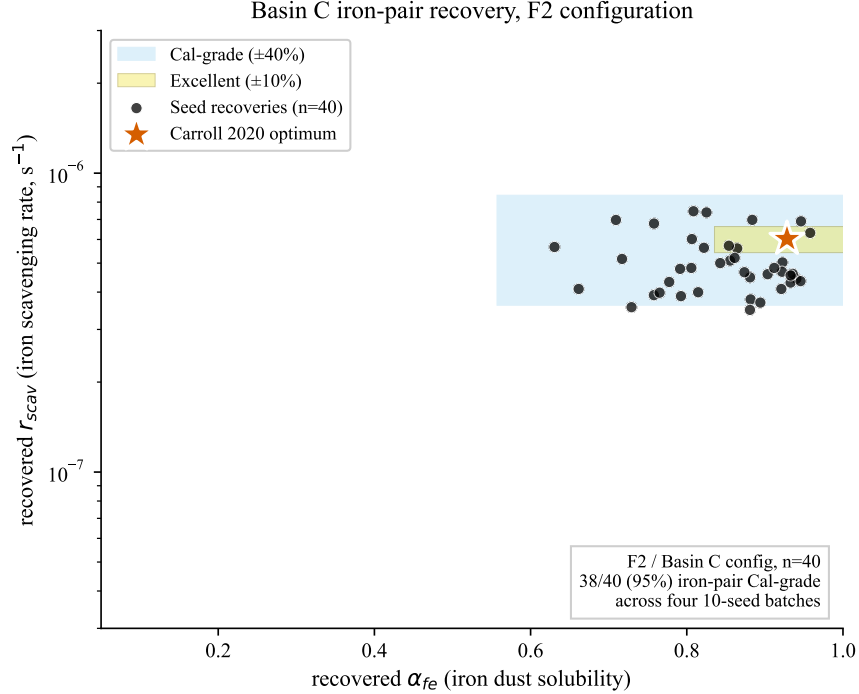


Figure 2: Iron-pair recovery at the F2 configuration ($\text{POSI_W}=1.0$, $\text{AOI_W_NATLSUBPOLAR}=2.0$, $\text{AOI_W_SOUTHERNOCEANPAC}=2.0$, $\text{CHL1_W_EXTRA}=3.0$) across $n=40$ seeds in four independent 10-seed batches. The red star marks Carroll 2020’s published optimum; shaded regions show $\pm 10\%$ (Excellent) and $\pm 40\%$ (Cal-grade) bounds. 38 of 40 seeds (95%) fall within the Cal-grade region for both `alpfe` and `scav_rat`.

(the small-cell phytoplankton parameters) rather than letting both target recoveries hold.

4.5 Binary mutex in observation anchors

Any non-zero PIC absolute anchor (`PIC_ABS_W`, tested down to 0.02 with paired POC anchors at multiple ratios) wipes iron-pair recovery to 0/10, regardless of magnitude. POC anchor alone also kills the iron pair, with a different downstream basin geometry. The mutex is *binary in PIC presence, not graduated in magnitude*. We interpret this as the PIC absolute anchor jointly constraining the product $R_{\text{PIC:POC}} \cdot M_{\text{tot}}$ via a tight basin, pulling the optimizer to a region where `R_PICPOC` is Cal-grade but the iron pair must drift to balance the loss.

This is the most actionable predictive finding for future observation-design choices: whether a candidate observation channel can coexist in the loss without inducing mutual mutex with iron recovery is a non-trivial question, and it is testable a priori by adding the channel to DarwinDiff’s loss and running a small sweep.

5 Discussion

5.1 The 5/6 ceiling as parameter conservation

Across 856 seeds in 86 single-lever configurations, no seed reaches 6/6 Cal-grade. The framing that fits the evidence is parameter conservation: the available observational constraints support

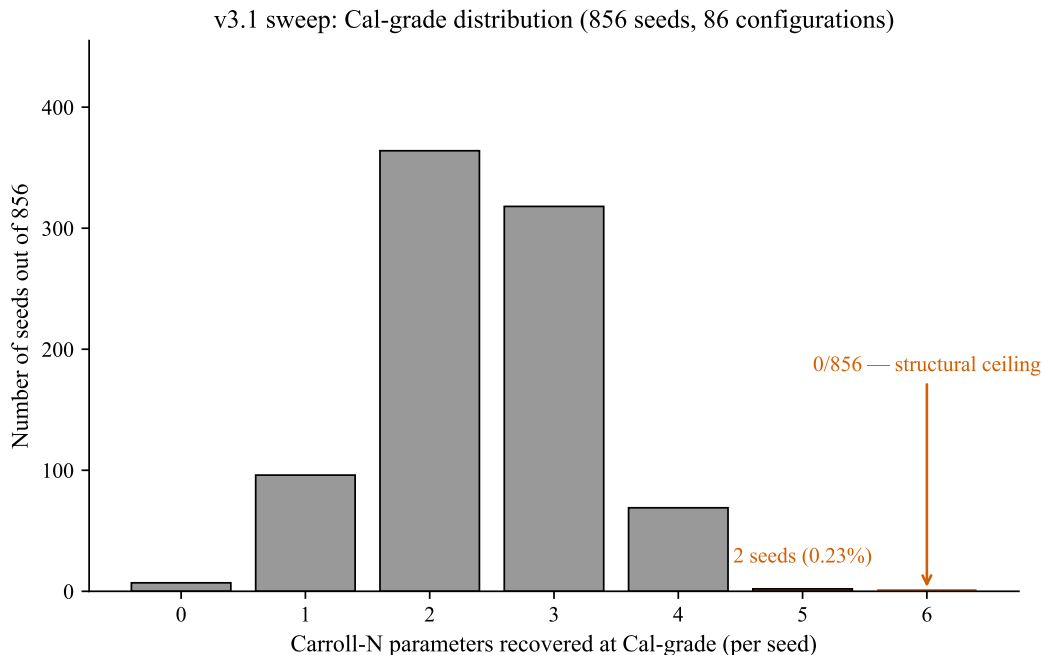


Figure 3: Distribution of Carroll-N parameters recovered at Cal-grade per seed, across 856 seeds in 86 single-lever configurations. Zero seeds reach all six (the structural 5/6 ceiling); only two seeds (0.24%) reach five, both unreproduced at $n=20$.

approximately five effective degrees of freedom over the 6-parameter Carroll-N surface, and the sixth parameter is always the residual sink — loss weighting decides which. The structural ceiling is supported by four independent pieces of evidence (§4.3), and the failure of the natural composition test (§4.4) is the strongest negative evidence: the two complementary 5/6 lever families do not compose, instead regressing below either parent.

This is informative for the broader question of which Carroll-N parameters are recoverable from ECCO-Darwin’s v05 observational basis. The 5/6 ceiling does not say “the method is broken”; it says “the current observational constraints are insufficient by one dimension.” Observation channels carrying information orthogonal to the current set — for example, satellite particulate inorganic carbon for the carbonate counter-pump, or additional GEOTRACES traverses constraining the iron pump — are the natural unblock. DarwinDiff’s framework makes such tests cheap: each new candidate channel is a loader and a loss weight.

5.2 Comparison with Green’s-functions calibration

Even with the structural ceiling, DarwinDiff exhibits substantial methodological advantages over Green’s-functions calibration:

- One gradient pass per training step replaces ~ 12 model runs per Gauss-Newton iteration. The 1500-epoch DINN training run on a single GPU completes in ~ 7 min, compared with multi-day full-model perturbation campaigns.
- New observation channels enter as additional loss terms rather than as a fresh Jacobian campaign. The marginal cost of adding a candidate channel is a loader and a loss weight.
- Per-cell parameter variation is strictly more expressive than the global-scalar class. The DINN class contains the global-scalar class as the constant-prediction limit, so DINN’s loss is at most

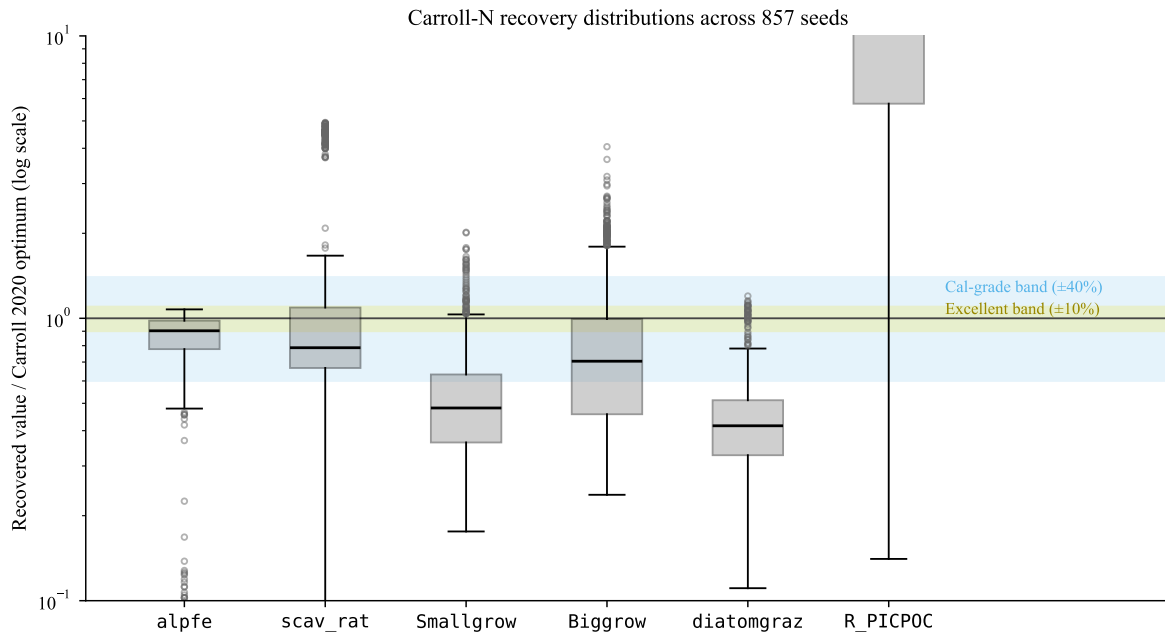


Figure 4: Distribution of recovered Carroll-N parameters across the v3.1 sweep, normalized by Carroll 2020’s published optima. Box plots show the IQR and whiskers at $1.5 \times \text{IQR}$ across 856 seeds; the shaded bands indicate Cal-grade ($\pm 40\%$) and Excellent ($\pm 10\%$). The iron pair (`alpfe`, `scav_rat`) clusters near unity; the other four parameters span broader distributions consistent with the parameter-conservation framing of the 5/6 ceiling.

equal to the optimal global-scalar fit, and typically lower whenever the target field has spatial structure expressible only by per-cell parameters.

5.3 Limitations

The current results are subject to several caveats.

- **Box-model proxy.** The 5–7-tracer box reduces the 39-tracer Darwin SMS kernel. The $5\text{-PFT} \rightarrow 2\text{-PFT}$ collapse is the dominant recovery-bias source in our companion experiments (Section 5.4). Eliminating the proxy via a port of `darwin_plankton.F` to differentiable PyTorch is the natural next step.
- **Per-cell architecture, no spatial coupling.** Real ocean BGC has advection and diffusion between cells, which the per-cell DINN ignores. This is appropriate for parameter recovery (the truth structure is per-cell) but not for a Track-2 emulator direction.
- **Climatology only.** All current fits use 23-year time-mean Darwin output. Time-resolved fitting opens a different scientific question (dynamical recovery, not steady-state) and is cluster-gated by GPU memory requirements.
- **DINNDeep does not extrapolate spatially.** Block cross-validation gives held-out Pearson $r=0.301$ versus in-distribution $r=1.000$ on iron, indicating that DINNDeep is fit-to-training rather than generalizing across spatial blocks. The DINN baseline is the appropriate variant for cross-AOI claims.

Table 3: Wave 6 composition test versus its two parent configurations, $n=10$ each.

Metric	Combined	w2e_peraoi_lam0.1	c_chl40_posi15
Seeds at 6/6	0/10	0/10	0/10
Seeds at 5/6	0/10	1/10	1/10
Seeds at $\geq 4/6$	0/10	1/10	1/10
Iron pair Cal-grade	9/10	10/10	10/10
R_PICPOC Cal-grade	0/10	0/10	1/10
diatomgraz Cal-grade	0/10	1/10	0/10
Mean Cal-grade count	2.00	2.40	2.70

5.4 Future work

Three priorities.

1. **Differentiable full Darwin, regional.** Port `darwin_plankton.F` to differentiable PyTorch, replacing the box-model proxy. Memory-feasible on existing single-GPU hardware (~ 15 GB for regional steady-state). Tests whether the 5/6 ceiling is structural-to-the-method or structural-to-the-proxy.
2. **Time-resolved fitting.** Multi-year dynamical recovery instead of climatological steady-state. Cluster-gated via cluster GPU memory; the relevant compute access is currently being pursued via MIT ORCD’s AICR (NVIDIA B200) beta program.
3. **Satellite observation channels.** Shelved for this paper to keep the v05 framing canonical; PACE 2024 ocean-color and MODIS particulate inorganic carbon loaders are built and parked for the leapfrog companion paper.

6 Conclusion

We adapt the BINN architecture (Xu et al., 2025) from terrestrial soil carbon to marine biogeochemistry, building a differentiable per-cell parameter recovery framework that replaces ECCO-Darwin’s Green’s-functions calibration with gradient-based inference. Across 856 seeds in 86 configurations of multi-AOI joint training, the framework recovers the Basin C iron pair reproducibly (38/40 Cal-grade) and exhibits a structural 5/6 ceiling on full Carroll-N recovery — supported by four independent pieces of evidence and consistent with a parameter-conservation framing. The ceiling is a diagnostic rather than a method failure: it predicts that observation channels orthogonal to the current set would unblock 6/6 recovery, and DarwinDiff’s framework makes such tests cheap. The methodological promise of differentiable parameter recovery for ocean biogeochemistry — gradient-cheap calibration with swappable observation channels — is delivered.

Code and data references

- Source code, sweep orchestration scripts, and per-seed result JSONs live in the project repository alongside this white paper (`src/darwindiff/`, `scripts/`, `D:\runs\bcr_*\`).
- ECCO-Darwin v05 outputs (the calibration target) are public via the ECCO Drive and the NASA GHG Center.
- GEOTRACES IDP2025 observations used in the loss are public via the GEOTRACES Data Assembly Centre.

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